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C303

1. Draft of Data Cleaning Documentation
2. Identify the Key Metrics and Measures in the chosen topics/dataset
3. Draft/Layout of the Dashboard

https://www.kaggle.com/datasets/aldinwhyudii/student-depression-and-lifestyle-100k-data/data

Draft of Data Cleaning Documentation (using C.L.E.A.N Framework)

Team Leader Note: I reviewed the datasets from Kaggle, and while it states that there are "no missing values", like it it's already "clean", we still need to verify this during our cleaning phase to ensure our 100,000 records from **Student Depression & Lifestyle** *is fully intact.*

Conceptualize (C)

- Defining the "grain" and the scope of the data to be cleaned.
- We are analyzing a survey of 100,000 records of every student involved in the said dataset. Each row captures the relationship between "Lifestyle" *(Sleep/Study/Social Media)* and the Mental Health "Outcome" *(Stress/Depression).*

Locate (L)

- Identifying specific areas in the dataset where errors or inconsistencies typically hide.
- We will specifically scan the **Department** and **Gender** columns. Since these are strings, we need to ensure there aren't multiple variations for the same category (e.g., "Eng" vs "Engineering"; "F" vs "Female") that would break our dashboard slicers.

Evaluate (E)

- Determining if outliers are "true data" or "errors" and deciding how to handle them.
- We will check the **Physical_Activity** and **Study_Hours** columns. If we ever find unrealistic values (like 168+ hours of study per week), we will flag them in an issues log to decide if they should be filtered or kept to "service the truth".

Augment (A)

- Enhancing the dataset with new calculations or "Clean" versions of columns.
- We will the create a new "Work-Life balance" metric by comparing Study_Hours vs Sleep_Duration. This will allow us to see if "grinding" correlates with the Depression flag more than the raw hours alone.

Note (N)

- Maintaining a transparent log of all data transformations for future audit.
- Our team will maintain a Transformation Log documenting every step, from our initial Student_ID duplicate check to our final outlier filters and to ensure our dashboard results are fully trustworthy.

Key Metrics and Measures (using Pyramid Framework)

Metrics (Foundation):
- CGPA (Float 0.0 - 4.0)
- Sleep_Duration and Study_Hours (both Float)
- Depression (Boolean: True/False)

KPIs (Signals):
- Depression Prevalence Rate:
  - (Count of Depression = True) / Total Student_ID) - this will tell us the overall mental health status of the 100,000 records of students.
  - Academic Performance Average: Average (CGPA) grouped by Department - and to identify if specific fields of study face higher academic pressure.
  - Lifestyle Impact Score: A correlation between Social_Media_Hours and Stress_Level (0 - 10).

OKR (North Star/ Goal):
- Our objective is to identify if lifestyle habits (Sleep and Physical Activity) can predict academic success/performance or depression risks to help university administrators drive better wellness policies.

Draft/Layout Dashboard (using DASH Framework)

Decision / Audience (D/A):
- Identifying which student subgroups require immediate wellness interventions.
- Our dashboard will be built for the *University* and *School Counselors*. This will then allow them to decide which specific Departments *(like Arts, Engineering, Medical, or Science)* need more mental health workshops based on real-time stress data.

Signal (S):
- Reducing the “noise” to focus on high-impact data points.
- We have chosen the **Depression Flag** and **Stress Level** *(0-10)* as our primary signals. By filtering out the non-essential data, we could ensure that the user will immediately see the “mental health temperature” of the campus.

Hierarchy (H):
- Using the “F-pattern” scan to guide the user from the big-picture totals to granular details.
- Layout:

- The Header(summary): Four high-level KPI cards showing **Total Students** *(100,000)*, **Average CGPA**, **Average Sleep Duration**, and **Overall Depression Rate**.
- The Middle(comparison): A clustered bar chart comparing **Department** vs **Average Stress Level.** This identifies the “hottest” areas of concern.
- The Footer(correlation):A scatter plot showing **Study Hours** vs. **CGPA** paired with a pie chart for **Gender Distribution** to help the users understand the ***“why”*** behind the numbers.